

GROSSEVICH, Russian Federation

WMO: 318230

Lat: 47.971N

Lon: 139.535E

Elev: 10

StdP: 14.69

Time Zone: 10.00 (E10)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-5.6	-21.2	1.7	-4.3	-18.3	2.0	-2.2	34.6	13.4	29.9	14.9	11.6	300	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	68.6	62.8	66.1	62.0	64.1	61.2	64.5	66.8	63.0	65.1	61.7	63.7	8.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.0	65.2	62.1	83.5	64.0	60.8	79.7	62.9	29.7	67.1	28.6	65.4	27.7	63.7	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
30.1	26.0	21.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-13.3	68.5	2.4	2.6	-15.0	70.4	-16.4	71.9	-17.7	73.4	-19.4	75.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	7.2	13.1	22.9	33.1	40.2	47.2	55.6	60.8	54.7	42.5	25.9	13.5
	DBStd	18.37	6.86	7.15	5.77	3.79	4.25	3.84	3.52	3.51	4.56	6.11	8.56	7.69
	HDD50	6221	1326	1032	841	507	307	101	3	0	12	239	723	1130
	HDD65	11014	1791	1452	1306	957	770	534	291	140	309	696	1173	1595
	CDD50	692	0	0	0	0	2	17	177	334	154	8	0	0
	CDD65	10	0	0	0	0	0	0	1	9	0	0	0	0
	CDH74	18	0	0	0	0	0	0	1	16	1	0	0	0
	CDH80	2	0	0	0	0	0	0	0	2	0	0	0	0
Wind	WSAvg	10.0	11.1	11.2	10.5	9.4	8.1	7.6	7.1	8.2	10.6	11.6	11.6	12.6
Precipitation	PrecAvg	38.5	1.6	1.3	2.4	2.6	3.3	3.8	5.1	5.7	4.0	3.4	3.0	2.4
	PrecMax	48.7	3.4	4.3	4.8	4.4	5.7	7.4	8.4	9.3	7.2	7.9	5.9	6.9
	PrecMin	27.2	0.4	0.1	0.8	0.8	1.2	1.0	1.4	2.7	1.5	0.8	0.8	0.6
	PrecStd	5.7	0.8	0.9	1.3	1.0	1.1	1.4	2.0	1.8	1.6	1.8	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	33.3	40.0	50.2	60.1	62.1	69.0	75.8	69.5	60.2	46.6	35.1
		MCWB	28.0	29.8	33.8	39.2	46.3	55.5	62.2	63.9	60.1	52.5	42.5	32.2
	2%	DB	25.8	29.7	35.0	44.0	52.9	57.4	63.8	70.3	66.1	57.0	43.0	31.0
		MCWB	23.3	26.4	30.4	36.6	45.0	53.6	61.2	64.4	60.7	51.6	38.6	28.2
	5%	DB	21.5	27.7	32.6	40.6	48.4	54.8	61.9	67.7	64.0	54.8	40.6	27.8
		MCWB	19.0	24.9	29.0	35.6	43.4	52.0	60.0	63.2	59.6	50.5	36.7	24.4
	10%	DB	17.8	24.8	31.0	38.4	45.8	52.7	60.3	65.6	62.2	52.8	38.0	24.6
		MCWB	15.4	21.4	27.9	35.2	42.3	50.8	58.9	62.4	58.3	48.8	33.8	21.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	30.4	34.7	41.4	48.8	56.8	64.3	68.1	64.4	55.8	43.2	32.1
		MCDB	29.6	33.4	38.1	46.0	54.5	60.3	66.1	71.3	66.5	58.1	45.0	33.6
	2%	WB	23.3	27.2	31.8	38.1	46.1	54.4	61.6	65.3	62.0	53.4	39.9	28.7
		MCDB	25.5	29.4	34.0	41.0	49.8	56.6	63.1	68.1	65.1	55.6	42.0	31.2
	5%	WB	18.9	25.0	30.1	36.8	44.1	52.3	60.3	64.1	60.4	51.4	37.0	24.7
		MCDB	21.2	27.7	31.7	39.2	47.1	54.2	61.6	66.4	62.9	53.8	40.0	27.7
	10%	WB	15.5	21.6	28.5	35.8	42.8	50.8	59.1	63.0	59.2	49.4	34.1	21.7
		MCDB	17.7	24.4	30.4	37.9	44.9	52.4	60.1	65.1	61.7	52.3	37.5	24.5
Mean Daily Temperature Range	5% DB	MDBR	15.0	14.7	11.9	8.2	7.6	6.5	5.8	7.6	11.6	13.8	14.2	14.1
		MCDBR	16.5	14.2	13.1	13.0	14.0	11.6	8.6	11.9	13.9	15.4	14.7	16.1
		MCWBR	14.7	12.2	10.5	8.8	8.4	8.1	6.4	7.7	10.0	12.5	12.8	13.9
	5% WB	MCDBR	16.1	13.5	11.2	10.5	11.0	9.8	7.8	9.8	10.5	13.5	13.3	14.8
	MCWBR	14.5	11.8	10.1	8.3	7.8	7.6	6.2	7.4	8.9	12.3	12.6	13.0	
Clear-Sky Solar Irradiance	taub	0.298	0.339	0.409	0.500	0.537	0.512	0.477	0.439	0.396	0.398	0.352	0.309	
	taud	2.213	2.088	1.953	1.834	1.843	2.020	2.187	2.308	2.347	2.167	2.168	2.175	
	Ebn at Noon	240	251	249	238	235	243	250	255	255	228	212	215	
	Edh at Noon	25	36	49	61	63	53	44	38	33	33	27	23	
All-Sky Solar Radiation	RadAvg	534	874	1232	1428	1544	1591	1493	1371	1188	814	532	417	
	RadStd	40	44	106	156	117	209	154	117	109	68	46	31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page